Egor Chulkov

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EU citizenship | open to relocation

Education

Skolkovo Institute of Science and Technology 2025 – Present
Ph.D. Student in Mathematics and Mechanics

Advisor: Prof. Oleg V. Vasilyev

Research Focus: Complexity-driven construction of sparse hierarchies for implicit Constructive Solid Geometry (CSG), combining per-voxel expression pruning based on interval arithmetic and localized Reverse Polish Notation (RPN) bit-masks with a threshold-based stopping rule that decouples memory and evaluation cost from scene complexity.

- Reduced memory on complex benchmark scenes by up to two orders of magnitude versus conventional dense subdivision (e.g., 107 MB to 0.16 MB on a scene of 5,487 primitives), with orders-of-magnitude faster evaluation than the unpruned expression.
- Compared pruning criteria within a single criterion-agnostic framework, showing that the interval-based Boolean criterion yields sparser hierarchies and lower construction overhead than Keeter- and Lipschitz-style value-preserving pruning.

Skolkovo Institute of Science and Technology 2023 – 2025
Master of Science in Mathematics and Computer Science *GPA: 5.0/5.0*

Thesis: *Adaptive Constructive Solid Geometry with constant evaluation complexity for modeling complex implicitly defined objects*

- Designed the foundational adaptive spatial octree methodology and C++ CPU ray-tracer that served as the architecture for subsequent PhD-level pruning optimizations.
- Benchmarked interval, affine, and revised affine arithmetic for expression pruning, implementing a custom interval-arithmetic library after finding that off-the-shelf libraries dominated the runtime and skewed the comparison.

Moscow Institute of Physics and Technology 2019 – 2023
Bachelor of Science in Applied Mathematics and Physics *GPA: 4.5/5.0*

Thesis: *Optimization of the characteristics of fiber optic distributed seismic sensors*

Publications & Preprints

- **Chulkov E.**, Vasilyev O. V., "Complexity-Driven Construction of Sparse Hierarchies for Implicit Constructive Solid Geometry." *ACM Transactions on Graphics*, 2026. **(Accepted with minor revisions)**
- **Chulkov E.**, Tikhotskiy S., Dubinya N., "Design of seismic sensors on DAS principle: analysis and numerical modeling," *GeoEurasia-2023*, Moscow.
- Tikhotskiy S., **Chulkov E.**, "On the possibility to design fiber optic seismic with the prescribed angular sensitivity diagram," *BalticPetroModel 2022*, St. Petersburg.

Patents

- **Chulkov E.**, Vasilyev O. V., "Computer-Aided Design System for Modeling Complex Implicitly Defined Objects." *Patent Application RU 2025131729*, Filed Nov 2025; decision to grant issued Oct 2026 after substantive examination. **(Granted, registration pending)**

- Nachev V., Novikova E., Vasilev I., **Chulkov E.**, Markov R., Sokolov V., Khudykin A., "Method and System for Early Detection and Prediction of Fire Spread in Forest Areas." *Patent RU 2 849 054 C1 (MIPT)*, Published Oct 2025.

Professional Experience

TESIS Summer 2024
Software Engineering Intern (Thermal Radiation Spectral Model) *Moscow, Russia*

- Engineered a C++ solver for computing equilibrium temperatures in convex enclosures.
- Implemented VRML file parsing functionality for geometry import.
- Implemented an analytical method based on Narayanaswamy's algorithm, improving accuracy up to 99%.

Sakhalin Energy 2023 – 2024
Research Engineer (Geomechanics Research Project) *Moscow, Russia*

- Developed a novel mathematical algorithm for calculating pore pressure field dynamics in producing wells.
- Engineered a calculation module to determine sand occurrence intensity based on well operation modes.
- Implemented inverse Laplace transform using Gaver-Stehfest algorithm for pressure field analysis.

Moscow Institute of Physics and Technology 2022 – 2024
Research Engineer, Laboratory of Wellbore Exploration *Moscow, Russia*

- Conducted fiber deformation modeling using ABAQUS finite element analysis software.
- Performed analytical studies on spirally twisted optical fiber deformation under plane wave influence.

Technical Skills

- **Computer Graphics:** F-rep, SDF, SVO, CSG, R-functions, OpenVDB
- **Scientific Computing:** C++, CUDA, OpenMPI, Python (NumPy, SciPy, Pandas), MATLAB
- **Simulation & Analysis:** ABAQUS, Wavelet Transform, Gaver-Stehfest Algorithm
- **Development Tools:** Git, Docker, Django
- **Languages:** English (B2/C1), Russian (Native), French (A1)

Honors and Awards

- T-Education Scholarship, Science track (2026)
- Academic Excellence Award, Skolkovo Institute of Science and Technology (2025)
- Prize winner of Phystech Olympiad (2018) and Phystech International Olympiad (2019)